

Project Design Phase Solution Architecture

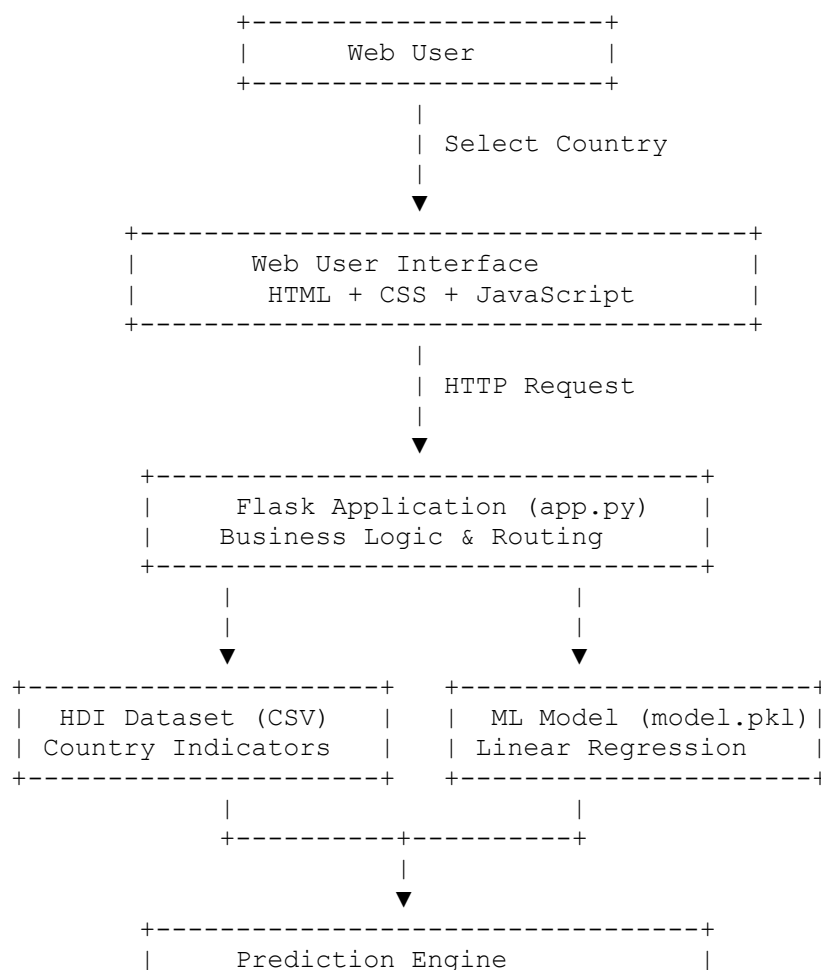
Date	18 JULY 2026
Team ID	
Project Name	A Comprehensive Measure of Well - being
Maximum Marks	4 Marks

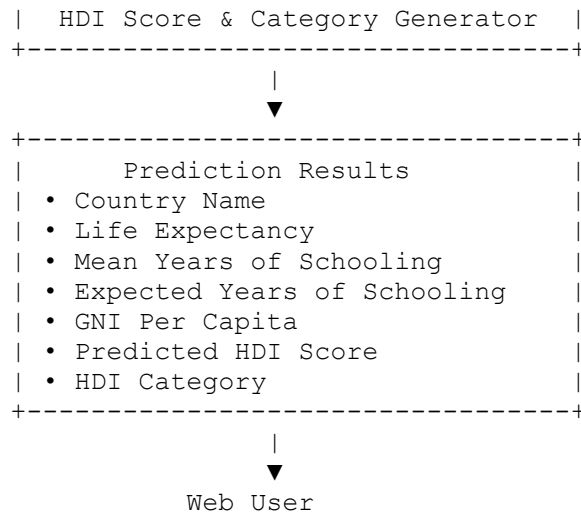
Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram





Components Used

Layer	Component	Technology
Presentation Layer	User Interface	HTML, CSS, JavaScript
Application Layer	Flask Application	Python, Flask
Machine Learning Layer	HDI Prediction	Scikit-learn (Linear Regression)
Data Layer	HDI Dataset	CSV File, Pandas
Model Storage	Trained Model	Joblib (model.pkl)

Solution Workflow

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graph TD; A[User Opens Website] --> B[Selects Country]; B --> C[Flask Receives Request]; C --> D[Reads Country Details from Dataset]; D --> E[Loads Trained ML Model]; E --> F[Predicts HDI Score]; F --> G[Determines HDI Category]; G --> H[Displays Results on Web Page];
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The solution workflow is a linear sequence of steps: 1. User Opens Website, 2. Selects Country, 3. Flask Receives Request, 4. Reads Country Details from Dataset, 5. Loads Trained ML Model, 6. Predicts HDI Score, 7. Determines HDI Category, 8. Displays Results on Web Page.

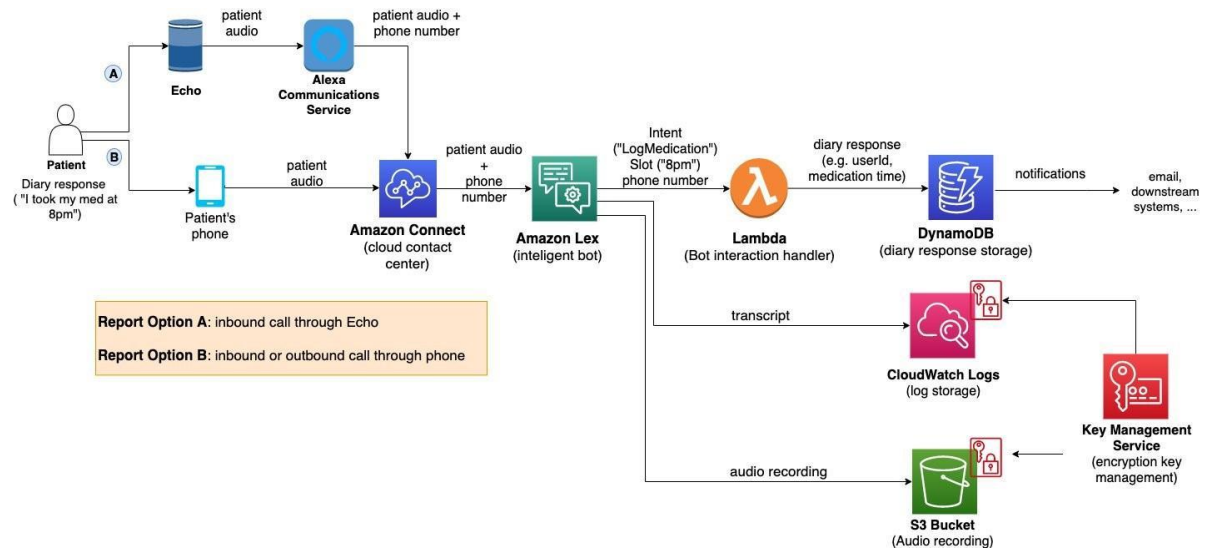


Figure 1: Architecture and data flow of the voice patient diary sample application

Reference: <https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>